Guidance, Normalization, and Looped Models

Why diffusion can guide, why per-token guidance is not sequence-level guidance, and what loops change

A companion note to X. Wu,
On the Emerging Connections Between Looped Models and Diffusion (October 2026)

October 20, 2026

The story in brief. Everything below follows from one fact,

a normalizing constant drops out only if it does not depend on the thing being sampled,

and one recurring idea: *applying a tilt one step at a time is myopic; exact tilting needs an expected future tilt.*

- **Diffusion** (§1). At each noise level the guided field is the score of a tilted density over the *entire* sample, so its normalizer is a constant and the gradient removes it. CFG is still myopic across noise levels, but that error vanishes as $t \to 0$ and no coordinate is ever frozen.
- **Per-token LLM guidance** (§2). The normalizer $C_w(y_{<t})$ depends on the prefix, which is part of the sample. The exact sequence-level conditional needs a lookahead factor V (a soft value function) that per-token guidance drops, and committed tokens are never revised.
- **Loops** (§3). A nearby loop reference makes each $\log C_w = O(\varepsilon^2)$, so the missing lookahead is second order. But $\log C_w = \mathrm{KL}(p_K \| q) + w\,\mathrm{KL}(p_K \| p_E)$ exactly, so at a fixed guided output only the second term shrinks. Refining a *whole-answer* state, rather than a next-token readout, is what would import diffusion's property in full.

Notation. Guidance strength $w \geq 0$. Tilting a strong distribution P_+ against a reference P_- means $P_+^{1+w} P_-^{-w} \propto P_+\,(P_+/P_-)^w$.

1 Why diffusion can do guidance

1.1 The normalizer is a constant

Let x_t be the noisy state at time t, produced from clean data by a forward kernel $k_t(x_t \mid x_0)$, for example $x_t = \alpha_t x_0 + \sigma_t \epsilon$. The key point is that x_t is the *entire sample*, every coordinate at once. Classifier-free guidance [1] combines a conditional and an unconditional score:

$$s_g = (1+w)\,\nabla_x \log p_t(x \mid c) - w\,\nabla_x \log p_t(x) = \nabla_x \log \tilde{\pi}_t(x),$$
$$\tilde{\pi}_t(x) = \frac{p_t(x \mid c)^{1+w}\, p_t(x)^{-w}}{Z_t}. \tag{1}$$

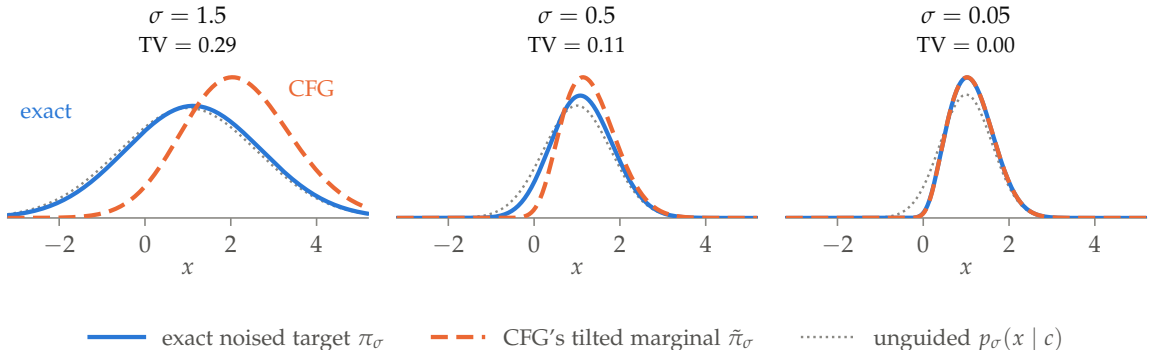


Figure 1: Exact noised target π_σ versus CFG's tilted marginal $\tilde{\pi}_\sigma$ for $p(x_0) = \frac{1}{2}\mathcal{N}(-1, 0.6^2) + \frac{1}{2}\mathcal{N}(1, 0.6^2)$, c = the right mode, $w = 3$, and $x_\sigma = x_0 + \sigma\epsilon$. At high noise CFG overshoots away from the other mode (mean 2.18 versus 1.13 at $\sigma = 1.5$). The total-variation gap falls to 0.0015 at $\sigma = 0.05$. The gap is the myopia of §1.2; its disappearance is fact 1 of §1.3.

The normalizer $Z_t = \int p_t(x \mid c)^{1+w} p_t(x)^{-w}\, dx$ depends on c, t and w, but not on x, so $\nabla_x \log Z_t = 0$. At every noise level, the guided field is the exact score of a normalized density over the whole sample, and score-based samplers (the reverse SDE, the probability-flow ODE, Langevin correctors) consume nothing but scores. The normalizer is never needed, for the same reason MCMC needs only an unnormalized density.

By Bayes' rule, $p_t(x \mid c)/p_t(x) = p_t(c \mid x)/p(c)$, so equivalently $\tilde{\pi}_t(x) \propto p_t(x \mid c)\, p_t(c \mid x)^w$: CFG is the conditional model plus w times an implicit classifier.

1.2 CFG is still myopic

The density we actually want at the end is the tilted data distribution

$$\pi_0(x_0) \;\propto\; p(x_0 \mid c)^{1+w}\, p(x_0)^{-w} \;\propto\; p(x_0 \mid c)\, p(c \mid x_0)^w.$$

Noising it exactly (Appendix A.1) gives

$$\pi_t(x_t) \;\propto\; p_t(x_t \mid c)\, h_t(x_t), \qquad h_t(x_t) = \mathbb{E}\big[\, p(c \mid x_0)^w \;\big|\; x_t,\, c \,\big], \tag{2}$$

whereas CFG's density is $\tilde{\pi}_t(x_t) \propto p_t(x_t \mid c)\, p_t(c \mid x_t)^w$ with $p_t(c \mid x_t) = \mathbb{E}[\, p(c \mid x_0) \mid x_t \,]$. The difference is where the power sits. The exact sampler weights each noisy state by the *expected tilt of the eventual clean sample*; CFG tilts by the classifier evaluated at the *current* noisy state. Since $\tilde{\pi}_t \neq \pi_t$ for $t > 0$, CFG does not sample π_0 exactly [3, 4].

1.3 Why diffusion gets away with it

Two structural facts rescue it.

1. **The future collapses.** As $t \to 0$, the posterior $p(x_0 \mid x_t, \cdot)$ concentrates at x_t, so $h_t(x)$ and $p_t(c \mid x)^w$ both tend to $p(c \mid x)^w$, and $\tilde{\pi}_t \to \pi_0$. The last steps of sampling use the exact tilted score, at any guidance strength.

2. **Nothing is frozen.** Every coordinate of x_t stays free until the end, so later steps (and corrector steps at a fixed t, which equilibrate to $\tilde{\pi}_t$ using only its score) can repair errors made at high noise. Bradley and Nakkiran make this precise by reading CFG as a predictor–corrector scheme [3].

Figure 1 shows both effects in one dimension.

2 Why per-token guidance is not per-sequence guidance

2.1 The normalizer depends on the prefix

Let $s = y_{<t}$ be the prefix and $p_\pm(\cdot \mid s)$ the next-token distributions of a strong and a reference predictor. Per-token guidance samples from

$$q(a \mid s) = \frac{r(a \mid s)}{C_w(s)}, \qquad r(a \mid s) = p_+(a \mid s)^{1+w}\, p_-(a \mid s)^{-w}, \qquad C_w(s) = \sum_a r(a \mid s). \tag{3}$$

Both models factorize autoregressively, so $\prod_t r(y_t \mid y_{<t}) = P_+(y)^{1+w} P_-(y)^{-w} =: R(y)$. Per-token sampling therefore produces $Q_{\text{local}}(y) = \prod_t q(y_t \mid y_{<t}) = R(y) / \prod_t C_w(y_{<t})$, while the per-sequence target is $Q^\star(y) = R(y)/Z$. Their ratio is

$$\frac{Q^\star(y)}{Q_{\text{local}}(y)} = \frac{\prod_t C_w(y_{<t})}{Z}. \tag{4}$$

The ratio is not constant, because C_w is a function of the prefix and the prefix is part of the sample. This is exactly the dependence that diffusion avoided: there, Z_t was a function of (c, t, w) alone. (This mismatch is the essay's Eq. 1 [10].)

2.2 The missing factor is an expected future tilt

The exact conditional of $Q^\star$ (Appendix A.2) is

$$\begin{aligned} Q^\star(a \mid s) \;&\propto\; r(a \mid s)\, V(s\,a), \\ V(y_{\le t}) &= \mathbb{E}_{y_{>t} \sim P_+}\Big[\big(\tfrac{P_+}{P_-}\big)^w (y_{>t} \mid y_{\le t})\Big] = \mathbb{E}_{y_{>t} \sim Q_{\text{local}}}\Big[\prod_{u>t} C_w(y_{<u})\Big]. \end{aligned} \tag{5}$$

V plays the role of h_t in (2): it is the expected tilt of the rest of the sequence, or equivalently the expected product of the normalizers still to come. Per-token guidance (3) is (5) with V dropped. It is exact only when $V(s\,a)$ is the same for every candidate a. That always holds at the last token, where $V \equiv 1$, and generally fails before it. Unlike diffusion, there is no way back: a committed token is never revisited.

The same object appears under three familiar names.

- **Soft value function.** $Q^\star = \arg\max_Q \mathbb{E}_Q[\log(P_+/P_-)] - \frac{1}{w}\mathrm{KL}(Q \| P_+)$ is a KL-regularized optimal policy, $\log V$ is its soft value function, and per-token guidance is the myopic,

zero-lookahead policy.

- **Label bias.** A locally normalized model cannot let later evidence redistribute mass among earlier choices, which is the label-bias problem that motivated globally normalized CRFs [5].
- **Learned lookahead.** Future discriminators [6] and twisted sequential Monte Carlo [7] are ways of approximating V.

2.3 A two-token example

Take $w = 1$ and two-token sequences. Both models are uniform over $\{\mathrm{A}, \mathrm{B}\}$ at step 1 and over $\{1, 2\}$ after A. After B the strong model is confident, $p_+(\cdot \mid \mathrm{B}) = (0.9, 0.1)$, while the reference is still uniform, $p_-(\cdot \mid \mathrm{B}) = (0.5, 0.5)$.

Table 1: Per-token versus per-sequence guidance in the two-token example ($w = 1$).

sequence y	$P_+(y)$	$P_-(y)$	$Q_{\text{local}}(y)$	$Q^\star(y)$
A1	0.25	0.25	0.250	0.189
A2	0.25	0.25	0.250	0.189
B1	0.45	0.25	0.494	0.614
B2	0.05	0.25	0.006	0.008
first token = B	0.50	0.50	**0.500**	**0.621**

At step 1 the two predictions agree exactly, so per-token guidance leaves the first token at 50/50. Sequence-level guidance still prefers B. Here $C_w(\mathrm{A}) = 1$ but $C_w(\mathrm{B}) = 0.9^2/0.5 + 0.1^2/0.5 = 1.64$, so $V(\mathrm{B}) = 1.64$, $V(\mathrm{A}) = 1$, and $Q^\star(\text{first} = \mathrm{B}) = 1.64/2.64 = 0.621$. Token B leads to a place where the strong model knows more than the reference, and per-token guidance cannot see that future.

3 What loops change

A looped model applies a shared block repeatedly before emitting each token. Reading out after E loops and after $K > E$ loops gives two predictions for the same prefix, p_E and p_K. Logit guidance, $z_g = z_K + w(z_K - z_E)$, is exactly the per-token rule (3) with $p_+ = p_K$ and $p_- = p_E$, because $\mathrm{softmax}\big((1+w)z_K - wz_E\big) \propto p_K^{1+w} p_E^{-w}$.[1] The earlier readout plays the role of a weaker version of the same model, as in autoguidance [2]. So everything in §2 applies. The question is what the loop structure buys.

[1] Hidden-state guidance, $h_g = h_K + w(h_K - h_E)$, reduces to this rule only when the readout is approximately affine along the intervention. Everything below is stated for the logit form.

3.1 An exact identity for the normalizer

Substituting $\log q = (1+w)\log p_K - w \log p_E - \log C_w$ into $\mathrm{KL}(p_K \| q)$ gives (Appendix A.3)

$$\log C_w(s) \;=\; \mathrm{KL}(p_K \,\|\, q) \;+\; w\,\mathrm{KL}(p_K \,\|\, p_E) \;=\; w\, D_{1+w}(p_K \,\|\, p_E), \tag{6}$$

where D_α is the Rényi divergence of order α. In words, the per-prefix normalizer is the size of the guidance step plus w times the distance between the two loop readouts. Both terms are non-negative, so $C_w \geq 1$, with equality only when the loops agree.

3.2 The essay's mechanism: close loops make the lookahead second order

For nearby loops, $z_K = z_E + \varepsilon u$, expanding (6) recovers the essay's Eq. 2:

$$\log C_w = \tfrac{1}{2}\, w(1+w)\, \varepsilon^2 \,\mathrm{Var}_{p_E}(u) + O(\varepsilon^3). \tag{7}$$

If $0 \leq \log C_w \leq \delta$ at every prefix, then $0 \leq \log V \leq (T-t)\,\delta$ (Appendix A.4). The missing lookahead in (5) therefore changes the log-odds between two candidates by at most $(T-t)\,\delta = O\big((T-t)\varepsilon^2\big)$, while guidance itself moves them by $w\varepsilon\,(u(a) - u(b)) = O(w\varepsilon)$. Per-token guidance then matches sequence-level guidance up to second-order corrections. This mirrors the first of diffusion's two rescues:

- CFG becomes exact as $t \to 0$ because the *future* collapses.
- Loop guidance becomes nearly exact as $\varepsilon \to 0$ because the *two predictions being contrasted* coincide.

3.3 The price: the floor does not move

The first term of (6) depends only on the guided distribution q, not on how the contrast that produced it was obtained. Hold q fixed and bring the reference closer, raising w to compensate. The second term then shrinks monotonically to zero (Appendix A.5), but the first does not:

$$\log C_w \;\geq\; \mathrm{KL}(p_K \,\|\, q) \qquad \text{for every reference and every } w. \tag{8}$$

Table 2 and Figure 2 run this on the example. The guided distribution after B is held at $q = (0.988,\, 0.012)$ while the reference moves from uniform toward p_K.

Table 2: The guided output q after B is held fixed while the loop reference moves toward $p_K = (0.9, 0.1)$.

reference $p_E(\cdot \mid \mathrm{B})$	w	$\mathrm{KL}(p_K \| q)$	$w\,\mathrm{KL}(p_K \| p_E)$	$C_w(\mathrm{B})$	$Q^\star(\text{first} = \mathrm{B})$
(0.50, 0.50)	1	0.127	0.368	1.640	0.621
(0.75, 0.25)	2	0.127	0.145	1.312	0.568
(0.85, 0.15)	4.75	0.127	0.052	1.195	0.544
(0.89, 0.11)	20.6	0.127	0.011	1.147	0.534
$\to$ (0.90, 0.10)	$\to \infty$	0.127	$\to 0$	$\to 1.135$	$\to 0.532$

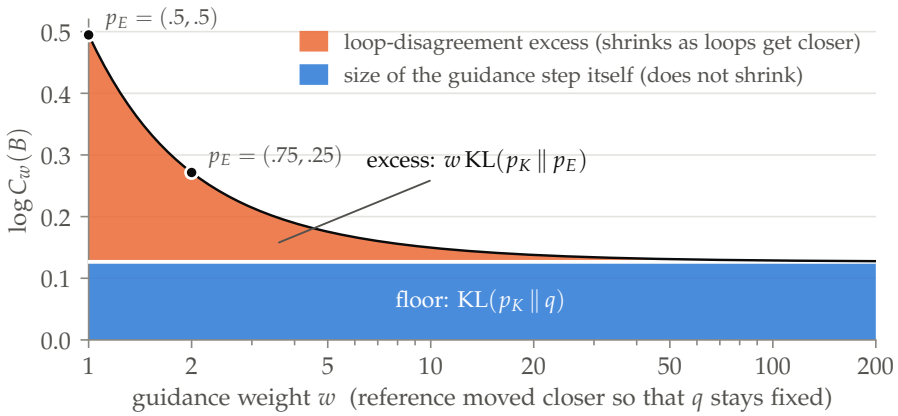


Figure 2: The identity (6) at a fixed guided distribution q after B. Moving the loop reference toward p_K (raising w to keep q fixed) removes the excess $w\,\mathrm{KL}(p_K\|p_E)$ but leaves the floor $\mathrm{KL}(p_K\|q) = 0.127$, which is the cost of moving the output at all. Dots mark the first two rows of Table 2.

Per-token guidance gives 0.500 for the first token in every row, so the gap to $Q^\star$ shrinks but never closes. Loops buy exactness only by making each per-token step gentle. Even then, the residual accumulates over T tokens, and nothing is revisited. Diffusion pays neither cost: at small t its proxy is exact at any guidance strength, and its state stays editable.

3.4 The full fix: refine a whole-answer state

The essay's Section 5 suggests training loop iterations as linked denoising steps. Corrupt a representation Y of the answer as $H = Y + \sigma\epsilon$; then the optimal denoiser satisfies Tweedie's formula [8, 9], $D^*(h) - h = \sigma^2 \nabla_h \log \rho_\sigma(h)$, so the loop's update is a score. What this buys depends on what Y represents.

- If Y represents the **whole answer**, guidance tilts a joint density over answers. Its normalizer depends only on the context, and the state stays revisable until the end, which are exactly the two properties behind §1. The remaining gap is diffusion's own, and it closes as $\sigma \to 0$.
- If Y represents only the **next token**, each position is tilted separately, and the product over positions picks up a per-prefix normalizer again, with the same structure as (4). Diffusing within one position does not fix the product across positions.

4 Summary

Table 3 collects the comparison.

Table 3: Where each form of guidance stands.

	object tilted	normalizer depends on	myopic error vanishes when	revisable
Diffusion CFG	whole sample	c, t only (constant in x)	$t \to 0$	yes
Per-token LLM guidance	next token	prefix (part of the sample)	last token only	no
Loop-contrast guidance	next token	prefix, but only at $O(\varepsilon^2)$	$\varepsilon \to 0$ (guidance weakens too)	no
Loop as denoiser of an answer state	whole answer	context only	$\sigma \to 0$	yes

Loop contrasts make per-token guidance a *controlled approximation* of sequence-level guidance. Only refining a whole-answer state makes it exact in the diffusion sense.

What to measure. The cleanest test of the essay's regime is direct. At the tokens where LoopCD changes the answer, estimate $\log V(s\,a)$ for the top few candidates by rollouts under Q_{local}. The per-token sampler already computes C_w at every step, so each rollout yields $\prod_u C_w$ at no extra model cost; average in log space (log-mean-exp) for stability. If $\log V$ varies little across candidates compared with the guidance shift, per-token guidance is effectively sequence-level where it matters. Reporting the two terms of (6) separately for the depth pairs used (the essay's Figure 4) would show how much of the normalizer the choice of depth pair actually controls.

A Derivations

A.1 Noising the tilted target

Using $p(x_0 \mid c)^{1+w} p(x_0)^{-w} = p(x_0 \mid c)\big[p(c \mid x_0)/p(c)\big]^w$,

$$\begin{aligned}\pi_t(x_t) &= \int \pi_0(x_0)\, k_t(x_t \mid x_0)\, dx_0 \;\propto\; \int p(x_0 \mid c)\, k_t(x_t \mid x_0)\, p(c \mid x_0)^w\, dx_0 \\ &= p_t(x_t \mid c) \int p(x_0 \mid x_t, c)\, p(c \mid x_0)^w\, dx_0,\end{aligned}$$

because $p(x_0 \mid c)\, k_t(x_t \mid x_0) = p(x_0, x_t \mid c) = p_t(x_t \mid c)\, p(x_0 \mid x_t, c)$ (the noise is independent of c). This is (2). CFG's form uses instead

$$\frac{p_t(x \mid c)}{p_t(x)} = \frac{p_t(c \mid x)}{p(c)}, \qquad p_t(c \mid x_t) = \int p(c \mid x_0)\, p(x_0 \mid x_t)\, dx_0.$$

A.2 The exact conditional of $Q^\star$

For fixed length T, define $V(y_{\le t}) = \sum_{y_{>t}} \prod_{u=t+1}^{T} r(y_u \mid y_{<u})$, with $V(y_{\le T}) = 1$. Summing R over completions gives $Q^\star(y_{\le t}) = \big[\prod_{u\le t} r(y_u \mid y_{<u})\big] V(y_{\le t})/Z$, hence

$$Q^\star(y_t \mid y_{<t}) = \frac{r(y_t \mid y_{<t})\, V(y_{\le t})}{V(y_{<t})}, \qquad V(y_{<t}) = \sum_a r(a \mid y_{<t})\, V(y_{<t}\, a), \qquad Z = V(\varnothing).$$

Writing $r = p_+ \,(p_+/p_-)^w$ gives the first form of V in (5); writing $r = q\, C_w$ gives the second. Q_{local} agrees with $Q^\star$ at step t if and only if $V(y_{<t}\, a)$ does not depend on a. The recursion for $\log V$ is the soft Bellman equation, and $Q^\star$ maximizes $\mathbb{E}_Q[\log(P_+/P_-)] - \frac{1}{w}\mathrm{KL}(Q\|P_+)$ by the Gibbs variational principle. With an end-of-sequence token, the same holds with $V = 1$ once the sequence has ended.

A.3 The identity (6) and the expansion (7)

Since $\log q = (1+w)\log p_K - w \log p_E - \log C_w$,

$$\mathrm{KL}(p_K\|q) = \sum_a p_K\big[\log p_K - \log q\big] = -w \sum_a p_K \log \frac{p_K}{p_E} + \log C_w = -w\,\mathrm{KL}(p_K\|p_E) + \log C_w.$$

For the Rényi form, $D_\alpha(P\|Q) = (\alpha-1)^{-1}\log\sum P^\alpha Q^{1-\alpha}$, and with $\alpha = 1+w$ the sum is C_w.

For the expansion, let $A(z) = \log\sum_a e^{z_a}$, so that $\log p = z - A(z)$ and

$$\log C_w = A\big(z_E + (1+w)\varepsilon u\big) - (1+w)\, A(z_E + \varepsilon u) + w\, A(z_E).$$

The constant terms cancel, $1-(1+w)+w = 0$, and so do the linear terms, $(1+w)-(1+w) = 0$. The quadratic term is $\frac{1}{2}\big[(1+w)^2 - (1+w)\big]\varepsilon^2\, u^\top \nabla^2 A\, u = \frac{1}{2} w(1+w)\varepsilon^2 \mathrm{Var}_{p_E}(u)$, because $\nabla^2 A = \mathrm{diag}(p_E) - p_E p_E^\top$.

A.4 The lookahead bound

By (6), $\log C_w \ge 0$. If also $\log C_w \le \delta$ at every prefix, then each of the $T-t$ future factors in $V(y_{\le t}) = \mathbb{E}_{Q_{\text{local}}}\big[\prod_{u>t} C_w(y_{<u})\big]$ lies in $[1, e^\delta]$. Hence $0 \le \log V(y_{\le t}) \le (T-t)\,\delta$, and $|\log V(s\,a) - \log V(s\,b)| \le (T-t)\,\delta$.

A.5 Fixed q: the excess shrinks, the floor stays

Fix p_K and the guided logits $z_g = z_K + \Gamma$. A reference that produces them with weight w has $z_E = z_K - \Gamma/w$, up to an additive constant. Writing $\mu = 1/w$,

$$w\,\mathrm{KL}(p_K\|p_E) = \langle p_K, \Gamma\rangle - \frac{A(z_K) - A(z_K - \mu\Gamma)}{\mu}.$$

The fraction is a backward difference quotient of the convex function A in direction Γ, so it is non-decreasing as $\mu \downarrow 0$ and converges to $\langle \nabla A(z_K), \Gamma\rangle = \langle p_K, \Gamma\rangle$. Hence the excess decreases monotonically to zero as $w \to \infty$, and $\log C_w \downarrow \mathrm{KL}(p_K\|q)$. For large w, the excess is approximately $\Gamma^\top \mathrm{Cov}_{p_K}\Gamma/(2w)$.

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